

■■ Express.js + Help Desk SaaS

A Beautiful Backend Learning Roadmap

55 Lessons

Express Fundamentals → Professional API → Multi-Tenant SaaS → Help Desk → Real-Time → Production

A structured roadmap for building your own professional Help Desk SaaS ■

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Phase 1 — Express Fundamentals

Build the foundation: request → middleware → controller → database → response.

Lesson 1 — Express server and first API

Learning goal: Master express server and first api and integrate it into the same evolving Help Desk SaaS project.

Lesson 2 — Routes and Controllers

Learning goal: Master routes and controllers and integrate it into the same evolving Help Desk SaaS project.

Lesson 3 — Request Body, POST, req.body

Learning goal: Master request body, post, req.body and integrate it into the same evolving Help Desk SaaS project.

Lesson 4 — PUT, PATCH, DELETE and HTTP status codes

Learning goal: Master put, patch, delete and http status codes and integrate it into the same evolving Help Desk SaaS project.

Lesson 5 — Middleware

Learning goal: Master middleware and integrate it into the same evolving Help Desk SaaS project.

Lesson 6 — Error handling

Learning goal: Master error handling and integrate it into the same evolving Help Desk SaaS project.

Lesson 7 — MySQL basics and database connection

Learning goal: Master mysql basics and database connection and integrate it into the same evolving Help Desk SaaS project.

Lesson 8 — CRUD with raw MySQL

Learning goal: Master crud with raw mysql and integrate it into the same evolving Help Desk SaaS project.

Lesson 9 — Prisma + MySQL setup

Learning goal: Master prisma + mysql setup and integrate it into the same evolving Help Desk SaaS project.

Lesson 10 — Prisma CRUD

Learning goal: Master prisma crud and integrate it into the same evolving Help Desk SaaS project.

■ YOU ARE HERE — NEXT LESSON

Prisma CRUD means learning the four core database operations: create, read, update, and delete.

```
const user = await prisma.user.create({
  data: { name: "Shohoz", email: "user@example.com" }
});
```

Phase 2 — Professional API Development

Turn basic endpoints into safer, authenticated, professional APIs.

Lesson 11 — Request validation

Learning goal: Master request validation and integrate it into the same evolving Help Desk SaaS project.

Lesson 12 — Password hashing with bcrypt

Learning goal: Master password hashing with bcrypt and integrate it into the same evolving Help Desk SaaS project.

Lesson 13 — User Registration

Learning goal: Master user registration and integrate it into the same evolving Help Desk SaaS project.

Lesson 14 — Login and JWT authentication

Learning goal: Master login and jwt authentication and integrate it into the same evolving Help Desk SaaS project.

Lesson 15 — Authentication middleware

Learning goal: Master authentication middleware and integrate it into the same evolving Help Desk SaaS project.

Lesson 16 — Roles and Permissions

Learning goal: Master roles and permissions and integrate it into the same evolving Help Desk SaaS project.

Phase 3 — Multi-Tenant SaaS Architecture

The heart of SaaS: every company must remain isolated from every other company.

Lesson 17 — Company model and database design

Learning goal: Master company model and database design and integrate it into the same evolving Help Desk SaaS project.

Lesson 18 — Company registration

Learning goal: Master company registration and integrate it into the same evolving Help Desk SaaS project.

Lesson 19 — User ↔ Company relationships

Learning goal: Master user ↔ company relationships and integrate it into the same evolving Help Desk SaaS project.

Lesson 20 — Tenant isolation

Learning goal: Master tenant isolation and integrate it into the same evolving Help Desk SaaS project.

Important SaaS rule:

```
Company A → Middleware identifies tenant → Query includes companyId = A  
Company B → Middleware identifies tenant → Query includes companyId = B
```

Lesson 21 — Company middleware

Learning goal: Master company middleware and integrate it into the same evolving Help Desk SaaS project.

Phase 4 — Help Desk Core Features

Transform the backend into a real Help Desk product.

Lesson 22 — Database design for tickets

Learning goal: Master database design for tickets and integrate it into the same evolving Help Desk SaaS project.

Lesson 23 — Create Ticket API

Learning goal: Master create ticket api and integrate it into the same evolving Help Desk SaaS project.

Lesson 24 — Ticket listing and filtering

Learning goal: Master ticket listing and filtering and integrate it into the same evolving Help Desk SaaS project.

Lesson 25 — Ticket details

Learning goal: Master ticket details and integrate it into the same evolving Help Desk SaaS project.

Lesson 26 — Ticket status and priority

Learning goal: Master ticket status and priority and integrate it into the same evolving Help Desk SaaS project.

Lesson 27 — Ticket assignment to agents

Learning goal: Master ticket assignment to agents and integrate it into the same evolving Help Desk SaaS project.

Lesson 28 — Departments

Learning goal: Master departments and integrate it into the same evolving Help Desk SaaS project.

Lesson 29 — Ticket messages / conversations

Learning goal: Master ticket messages / conversations and integrate it into the same evolving Help Desk SaaS project.

Lesson 30 — Internal notes

Learning goal: Master internal notes and integrate it into the same evolving Help Desk SaaS project.

Phase 5 — Advanced Features

Add scale, security, automation, search, queues, and performance.

Lesson 31 — File uploads

***Learning goal:** Master file uploads and integrate it into the same evolving Help Desk SaaS project.*

Lesson 32 — Secure file storage

***Learning goal:** Master secure file storage and integrate it into the same evolving Help Desk SaaS project.*

Lesson 33 — Pagination

***Learning goal:** Master pagination and integrate it into the same evolving Help Desk SaaS project.*

Lesson 34 — Search

***Learning goal:** Master search and integrate it into the same evolving Help Desk SaaS project.*

Lesson 35 — Sorting and filtering

***Learning goal:** Master sorting and filtering and integrate it into the same evolving Help Desk SaaS project.*

Lesson 36 — Email notifications

***Learning goal:** Master email notifications and integrate it into the same evolving Help Desk SaaS project.*

Lesson 37 — Notification system

***Learning goal:** Master notification system and integrate it into the same evolving Help Desk SaaS project.*

Lesson 38 — Redis

***Learning goal:** Master redis and integrate it into the same evolving Help Desk SaaS project.*

Lesson 39 — Queues and background jobs

***Learning goal:** Master queues and background jobs and integrate it into the same evolving Help Desk SaaS project.*

Lesson 40 — Rate limiting and API security

Learning goal: Master rate limiting and api security and integrate it into the same evolving Help Desk SaaS project.

Phase 6 — Real-Time Features

Move into live, event-driven application features.

Lesson 41 — Socket.IO introduction

Learning goal: Master socket.io introduction and integrate it into the same evolving Help Desk SaaS project.

Lesson 42 — Real-time ticket updates

Learning goal: Master real-time ticket updates and integrate it into the same evolving Help Desk SaaS project.

Lesson 43 — Real-time notifications

Learning goal: Master real-time notifications and integrate it into the same evolving Help Desk SaaS project.

Lesson 44 — Online agent status

Learning goal: Master online agent status and integrate it into the same evolving Help Desk SaaS project.

Phase 7 — Professional Backend

Apply professional backend engineering practices.

Lesson 45 — Logging

Learning goal: Master logging and integrate it into the same evolving Help Desk SaaS project.

Lesson 46 — API response structure

Learning goal: Master api response structure and integrate it into the same evolving Help Desk SaaS project.

Lesson 47 — Environment configuration

Learning goal: Master environment configuration and integrate it into the same evolving Help Desk SaaS project.

Lesson 48 — API documentation

Learning goal: Master api documentation and integrate it into the same evolving Help Desk SaaS project.

Lesson 49 — Testing

Learning goal: Master testing and integrate it into the same evolving Help Desk SaaS project.

Lesson 50 — Security improvements

Learning goal: Master security improvements and integrate it into the same evolving Help Desk SaaS project.

Phase 8 — Deployment

Take the application from local development to production.

Lesson 51 — Production environment setup

Learning goal: Master production environment setup and integrate it into the same evolving Help Desk SaaS project.

Lesson 52 — PM2

Learning goal: Master pm2 and integrate it into the same evolving Help Desk SaaS project.

Lesson 53 — Nginx

Learning goal: Master nginx and integrate it into the same evolving Help Desk SaaS project.

Lesson 54 — Production deployment

Learning goal: Master production deployment and integrate it into the same evolving Help Desk SaaS project.

Lesson 55 — Monitoring and maintenance

Learning goal: Master monitoring and maintenance and integrate it into the same evolving Help Desk SaaS project.

Final Architecture Blueprint

By Lesson 55, your application should connect these systems together:

Client / Frontend ↓ Express Routes ↓ Middleware (Auth → Company/Tenant → Validation → Rate Limit) ↓ Controllers / Services ↓ Prisma → MySQL Redis → Cache / Queue support Socket.IO → Real-time events Background Jobs → Email & slow processing PM2 + Nginx → Production Logging + Monitoring → Maintenance

■ *Study rule: do not jump randomly. Build each lesson on top of the previous lesson until one complete Help Desk SaaS application exists.*

Happy Building ■